

Yuchen Wang

Passionate AI research engineer, striving to solve challenging real-world problems.

LINKS

Email: yuchenw@stanford.edu
Page: <https://yuchenwyc.com>
Github:// [yuchenWYC](#)
LinkedIn:// [yuchenWYC](#)

EDUCATION

Stanford University

SEP 2021 - MAR 2023
Master of Science in Computer Science
Specialization: Human-Computer Interaction

University of Toronto

SEP 2017 - JUN 2021
Honours Bachelor of Science
Computer Science & Statistics,
minor in Mathematics
cGPA: 3.98/4, Course Avg.: A+

AWARDS

(Jan 21) Konrad Group Women in Technology Scholarship \$2000
(Jan 21) The Dorothy Walters Scholarship \$2000
(May 20) UofToronto Excellence Award \$6,000
(Dec 19) The Dorothy Walters Scholarship \$600
(All years) Dean's List Scholar

SKILLS

Specialized

Python • C • Unix/Linux • $\text{\LaTeX}$ •
Git • TypeScript • ReactTS •
HTML & CSS

Familiar

R • C++ • MatLab • SQL • Java
(Android)

Machine Learning Libraries

NumPy • PyTorch • Pandas •
Matplotlib • ggplot2 • SciPy •
Scikit-learn • LangChain

WORK EXPERIENCE

Microsoft Bing Search | SOFTWARE ENGINEER

June 2023 - July 2024 | Redmond, WA

- Designed and implemented a logging system for large language model product pipelines that is reliable, easily readable and easy to query.
- Designed and supported features for Azure Semantic Search, engineer solid production workflows for new cases.
- Technologies: C#, Kusto

Microsoft Bing Search | APPLIED AND DATA SCIENTIST INTERN

June 2022 - September 2022 | Mountain View, CA

- Generated multiple levels of negative data for language retrieval models using Approximate Nearest Neighbor on Euclidean distance of encoded query vectors.
- Designed new evaluation metrics that are universal for all natural languages (including URL overlap rate and modified BLEU/Rouge scores) to calculate relevance scores among search queries and docs.
- Developed a pipeline that effectively evaluates query-doc relevance scores using Transformer-based models.
- Technologies: Python, PyTorch, Pandas, SQL

Ernst & Young | TECHNICAL CONSULTANT INTERN (SOFTWARE)

June 2021 - August 2021 | Shanghai, China

- Developed a software pipeline to help an aviation authority extract formatted data from real-time video streams.
- Applied optical character recognition deep learning models on the client's datasets.
- Communicated closely with the client, wrote detailed documentations.
- Technologies: Python, OpenCV2, Pandas, Docker

Vector Institute | MACHINE LEARNING INTERN - PROF. ROGER GROSSE

April 2020 - January 2021 | Toronto, ON, Canada

- Designed and implemented hypernetwork algorithms (Self-Tuning Networks) to auto-tune hyperparameters of neural networks during a single run.
- Applied the algorithm on training Bayesian Neural Networks. The experimental results showed that the new architecture helps solving the difficult posterior collapse problem.
- Proven the related mathematical theories.
- Technologies: Python, NumPy, PyTorch, Autograd, Slurm

University Health Network | MACHINE LEARNING INTERN - PROF. BO WANG

May 2019 - December 2019 | Toronto, ON, Canada

- Worked with clinicians from the Toronto General Hospital to design a data processing pipeline for electronic health records time-series.
- Engineered modular code for RNNs, Transformers and tree-based models.
- The final models accurately predict one-year and two-year outlook cause of death for post-organ-transplant patients with 0.7-0.8 auc-roc.
- Technologies: R, Pandas, Python, NumPy, PyTorch, Scikit-learn

PROJECTS

Project X Research Competition | September 2020 - November 2020

- Led the UofT team to develop a neural ordinary differential equation architecture, which learns the dynamics of time series data with multiple predictors.
- Applied the model on semi-synthetic plant disease datasets and achieved remarkable results in both extrapolation and interpolation.
- Was awarded a \$20,000 prize as the competition winner. [[ICML 2021 workshop](#)]